

OpsDocs

DailyOps Document Operations System

Product Requirements & System Architecture v1.0

CREATE Structured creation	CONTROL Review · approval · signatures · audit
PROVE Issued artifacts · integrity · verification	ARCHIVE Searchable system of record

Purpose

Formalize the product so the project can move directly into UX, data modelling, architecture, implementation, testing and launch.

01 Executive definition

OpsDocs is the internal **DailyOps Document Operations System**. It converts the DailyOps Document System into an executable platform: select a controlled document type, enter structured data, generate the document, route it through review and approval, apply signatures, preserve evidence, issue the final artifact and archive it.

The central abstraction is **Template** → **Data** → **Document Instance** → **Workflow** → **Evidence** → **Archive**. OpsDocs is therefore not a CRM, not a generic word processor and not a folder containing Word templates.

The document is treated as a **controlled business record**, not merely as a file.

02 Product vision and objectives

Vision: one controlled system for producing, reviewing, approving, signing, issuing, verifying and retrieving every official, business, legal/formal and certificate document issued by DailyOps.Tech.

Objectives: eliminate template proliferation; standardize identity and metadata; reduce manual re-entry; prevent numbering collisions; make responsibility visible; preserve a complete audit history; make signatures explicit; generate consistent PDF/DOCX outputs; support certificate verification; provide a defensible record of document history.

Success condition: an authorized DailyOps user can produce a correct document without knowing how the underlying layout or rendering system works.

03 The 13 product pillars

1. **Document Library** — controlled catalogue of families and document types.
2. **Document Builder** — guided forms instead of free-form layout editing.
3. **Template Engine** — versioned templates, placeholders, conditions, reusable blocks and signatures.
4. **Document Numbering** — central, transactional references such as DO-BIZ-2026-0042.
5. **Version & Revision Control** — separate template versions, document revisions and final artifacts.
6. **Review & Approval Workflow** — explicit states, roles, decisions and rejection/change reasons.
7. **Signature System** — Prepared by, Reviewed by, Approved by, Authorized by, Signed by, Accepted by.
8. **Document Inbox** — operational queues for drafts, review, approval, signature, changes and archive.
9. **Audit Trail** — append-only evidence of who did what, when, to which revision and with which result.
10. **Controlled Data Reuse** — clients, contacts, projects, services and company data reused safely.
11. **Output & Distribution** — authoritative PDF, optional DOCX, controlled download/share.
12. **Certificate & Verification Layer** — certificate IDs, QR verification, validity and revocation.
13. **Integration & Platform Layer** — CRM, accounting, email, storage, identity and e-signature integrations.

04 Document taxonomy

OFFICIAL — DailyOps.Tech speaks. Communiqués, announcements, official letters, notices, clarifications, denials, confirmations, minutes and institutional correspondence.

BUSINESS — DailyOps.Tech delivers. Proposals, offers, quotations, SOWs, invoices, intervention reports, assessment reports, deliverables and client guides.

LEGAL / FORMAL — DailyOps.Tech commits. Contracts, NDAs, agreements, engagement letters, amendments, terms, SLA/DPA documents and contractual notices.

CERTIFICATE — DailyOps.Tech recognizes. Training, internship, participation, contribution and recognition certificates.

Each type receives a unique machine identifier, e.g. **DO-BIZ-PROPOSAL**, **DO-LEG-NDA**, **DO-OFF-LETTER**, **DO-CER-INTERNSHIP**.

05 Core user journey

1 Open Inbox → **2** New Document → **3** Select family → **4** Select type → **5** Load client/project data → **6** Complete form → **7** Preview → **8** Save draft → **9** Submit for review → **10** Review/change request → **11** Approve → **12** Generate final → **13** Sign → **14** Issue → **15** Archive.

Every meaningful transition creates an audit event containing actor, timestamp, previous state, new state and relevant revision/artifact information.

06 Document lifecycle and state machine

Recommended lifecycle: **Draft** → **In Review** → **Changes Requested** → **Approved** → **Ready to Sign** → **Signed** → **Issued** → **Archived**.

Optional states: **Rejected**, **Cancelled**, **Expired**, **Revoked**, **Superseded**.

Users cannot skip mandatory workflow steps. A signed/issued artifact is immutable. Corrections after signature create a new revision or replacement record according to the document policy.

Each state transition requires a system event. Sensitive transitions may require a reason, comment or evidence attachment.

07 Document Inbox

The Inbox is the operational control surface. Its primary question is: **What needs my attention?**

Queues: **My Drafts**, **Needs My Review**, **Needs My Approval**, **Needs My Signature**, **Changes Requested**, **Recently Issued**, **Expiring / Renewal**, **Certificates**, **Archived**.

Each item shows document number, title, type, client/recipient, owner, state, revision, pending action, age, due date and last activity.

Contextual actions: Open, Review, Approve, Request Changes, Sign, Generate, Download, Share, Archive, Revoke or Cancel.

Dashboard counters must be operational: for example **4 awaiting review**, **2 awaiting signature**, **1 overdue**.

08 Dynamic Document Builder

The builder is form-driven. Users do not manipulate page geometry.

A document type definition specifies field name, label, type, required/optional state, validation, source, default, visibility condition, repeatability, formatting and help text.

Field types: text, rich text, date, date range, number, currency, percentage, address, person, organization, contact, project, service, line-item table, checkbox, select, multi-select, attachment, calculated value and signature role.

Conditional logic can show tax fields only when applicable, add signatories for sensitive documents, or expose programme-specific certificate fields.

Required capabilities: autosave, validation, preview, draft recovery and change tracking.

09 Template engine

Templates are controlled system assets, not ordinary files copied between folders.

A template contains layout, reusable components, field bindings, conditions, typography, header/footer rules, numbering rules, signature blocks, page rules and output configuration.

Reusable components should include DailyOps header, metadata block, address block, subject block, standard clauses, commercial tables, legal signature block, certificate block, confidentiality footer and page footer.

Template versioning is independent from document revision. Example: Proposal template v1.3 can generate DO-BIZ-2026-0042 revision 4.

Only approved template versions may issue final documents. Retired versions remain renderable for historical records but cannot be selected for new documents.

10 Numbering and document identity

References are system-generated and immutable. Proposed pattern: **DO-[FAMILY]-[YEAR]-[SEQUENCE]**.

Examples: DO-OFF-2026-0042, DO-BIZ-2026-0043, DO-LEG-2026-0044, DO-CER-2026-0045.

Allocation must be transactional to prevent collisions under concurrent use.

Every document also receives an internal UUID. The human reference is for business use; the UUID is the technical identifier.

Identity metadata includes type code, title, status, owner, recipient/client, issue date, effective date when relevant, template version, confidentiality and lifecycle state.

11 Versioning and revision model

Separate four concepts: **Template Version**, **Document Revision**, **Generated Artifact**, **Signed/Issued Record**.

A draft can be revised without changing its document number. A PDF is an artifact of a specific revision. Once issued, that artifact becomes the authoritative record.

Example: DO-BIZ-2026-0042 Rev.0 → Rev.1 → Rev.2 → approved → signed artifact → issued. A later contractual change should follow the document's amendment/replacement rules rather than silently altering the issued record.

The UI should show a revision timeline and a change summary between revisions.

12 Review and approval

Review and approval are different controls. A reviewer checks correctness, completeness, content and internal compliance. An approver exercises authority to authorize issuance.

Suggested roles: **Prepared by**, **Reviewed by**, **Approved by**, **Authorized by**, **Signed by**, **Accepted by**.

Each document type defines which roles are mandatory, optional or not applicable.

Example proposal workflow: Prepared by → Reviewed by → Approved by → Issued. Example NDA: Prepared by → authorized review → Authorized by → signatures.

Reviewer actions: approve review, request changes, comment, attach evidence. Approver actions: approve, reject, request changes. Every decision is auditable.

13 Signature system

Signatures are workflow events and evidence, not decorative text blocks.

Signature roles are independent from their visual representation. A signature may render as name/title/date, uploaded signature image, integrated e-signature, or a combination.

Minimum signature metadata: signer identity, role, timestamp, authentication context where available, document revision, artifact hash and signature status.

Each issued artifact should have a cryptographic hash, preferably SHA-256, stored with the artifact and signature/issuance event.

For external signatures, integrate a dedicated e-signature provider in a later phase rather than inventing a legally sufficient signature mechanism in the MVP.

14 Audit trail and evidence model

The audit trail is a first-class subsystem. It must answer: **who did what, to which document, when, from which state, resulting in which state, and with what evidence?**

Events include: created, field_updated, attachment_added/removed, draft_saved, revision_created, submitted_for_review, review_started, review_approved, changes_requested, approval_granted/rejected, generated, regenerated, signature_requested/completed, issued, downloaded, shared, archived, restored, cancelled, revoked, superseded and template_changed.

Each event contains: event ID, document ID, actor ID, actor role, event type, timestamp, source/application, previous state, new state, revision, change summary, reason/comment, related artifact and integrity metadata.

Audit events are append-only. Ordinary users cannot edit or delete them. Retention rules are explicit and themselves governed.

The document workspace exposes a readable activity timeline; authorized administrators can inspect structured event records.

Event	Evidence
approval_granted	actor · timestamp · revision · state transition · comment · artifact
generated	template version · revision · artifact ID · SHA-256
signature_completed	signer · role · timestamp · artifact hash · status

15 Roles, permissions and access control

Use RBAC with document-level and action-level permissions.

Suggested roles: **Administrator, Document Manager, Author, Reviewer, Approver, Signer, Viewer/Auditor.**

Permissions include create, edit, view, submit, review, approve, sign, issue, download, share, archive, revoke, manage templates, manage numbering, view audit trail and administer users.

Separation of duties should be enforceable for sensitive documents, preventing one person from both preparing and giving final approval where policy requires.

Confidentiality levels can be applied to documents and inherited by artifacts.

16 Core data model

Core entities: **User, Organization, Contact, Client, Project, DocumentType, Template, TemplateVersion, Document, DocumentRevision, DocumentFieldValue, Artifact, WorkflowDefinition, WorkflowInstance, WorkflowAction, SignatureRequest, SignatureEvent, AuditEvent, Attachment, Certificate, VerificationRecord.**

Relationships: Document belongs to DocumentType; Document uses TemplateVersion; Document has many Revisions; a Revision generates Artifacts; WorkflowInstance governs a Document; AuditEvents reference Document and optionally Revision/Artifact; Certificates reference a Document and expose verification.

Business entities are reusable. A client is created once and referenced by proposals, contracts, invoices and reports.

Structured field values must be stored separately from rendered output so documents remain searchable, regenerable and integrable.

17 Technical architecture

Frontend: Next.js / React with form engine and live document preview.

API/application: Node.js + TypeScript, modular domains for documents, templates, workflow, signatures, audit and integrations.

Database: PostgreSQL for transactional data, structured fields and workflow state.

Object storage: S3-compatible storage for PDFs, DOCX files, attachments and signature artifacts.

Rendering: HTML/CSS → PDF as the controlled rendering path; DOCX export when editing is explicitly required. Rendering must be deterministic and versioned.

Authentication: organization identity provider or secure local auth initially; MFA for privileged users.

Background jobs: PDF generation, notifications, QR generation, hashing and integration tasks.

Observability: application logs, metrics and job status remain conceptually separate from the business audit trail.

18 API and integration strategy

OpsDocs should be API-first even if the first interface is internal.

Initial API domains: **/documents, /document-types, /templates, /template-versions, /clients, /projects, /workflows, /approvals, /signatures, /artifacts, /audit-events, /certificates, /verification.**

Future integrations: CRM for customer/project data; accounting for invoices; email for distribution; storage for archive; identity provider for users; e-signature for external signing; automation systems for notifications.

External systems may provide data, but OpsDocs remains the system of record for the document lifecycle and its audit trail.

19 Certificate and verification layer

Certificates need a specialized flow because recipients may need to prove authenticity without internal access.

Each certificate receives a unique certificate ID and optional QR verification URL, for example **verify.dailyops.tech/DO-CER-2026-00482.**

Public verification exposes only appropriate data: validity, certificate number, holder if authorized, programme/title, issue date, issuer and status. Internal workflow data remains private.

Statuses: **Valid, Revoked, Expired, Superseded, Not Found.**

The verification page confirms authenticity without exposing the internal audit trail.

20 Security and integrity requirements

Documents can contain personal, commercial and contractual information. Security is therefore a core architecture concern.

Baseline controls: encrypted transport, encrypted storage where appropriate, least privilege, MFA for privileged users, secure sessions, protected secrets, backups, attachment controls, access logging, rate limiting, secure downloads and explicit retention policies.

Artifact integrity: calculate SHA-256 for issued artifacts and store it with artifact and issuance/signature records.

Template publication, template retirement, numbering configuration, role changes and audit access must themselves be auditable.

21 Document generation and output rules

PDF is the authoritative issued format. DOCX is an editable convenience format and must not silently replace the issued PDF.

Generation is reproducible from document revision + template version + data snapshot. The system retains enough data to explain how an artifact was generated.

Pre-issue checks: missing required fields, page count, overflow/layout warnings, signature completeness, numbering, footer metadata and file integrity.

Final PDF metadata should contain document number, title, issuer, issue date, revision and relevant application identifiers.

22 Notifications and automation

Notifications are event-driven: submitted for review, changes requested, approval granted, signature requested/completed, issued, certificate nearing expiry and workflow overdue.

Start with in-app notifications and email. Other channels can be integrations later.

Notifications link to the work item and respect permissions.

Scheduled jobs may surface overdue items without changing document state automatically unless a policy explicitly permits it.

23 Search, retrieval and reporting

Search is based on structured metadata, not only filenames.

Filters: family, type, reference, status, client, project, owner, date range, reviewer, approver, signer, confidentiality, template version and certificate status.

Full-text indexing of generated content can be added later; structured fields remain the reporting source of truth.

Reports: documents issued, pending actions, approval cycle time, revisions per document, template usage, certificate status, overdue workflows and audit activity.

24 MVP scope and phases

Phase 1 — Foundation: authentication, users/roles, document types, templates, numbering, document creation, draft saving, PDF generation, Inbox and basic audit trail.

Phase 2 — Control: review/approval workflow, comments, revision history, role separation, signatures as controlled events, artifact hashes and archive.

Phase 3 — Business data: clients, contacts, projects, services, reusable data and commercial calculations.

Phase 4 — Certificates: certificate issuance, QR verification, validity and revocation.

Phase 5 — Integrations: CRM, accounting, email, storage, e-signature, API and webhooks.

Do not build the entire CRM, accounting platform or e-signature provider inside OpsDocs.

25 Initial document types

OFFICIAL: Official Letter, Official Notice/Announcement, Information Note, Administrative Confirmation.

BUSINESS: Service Proposal, Quotation, Invoice, Intervention Report, Service Report.

LEGAL/FORMAL: NDA, Service Agreement, Engagement Letter, Amendment.

CERTIFICATE: Internship Certificate, Training/Participation Certificate, Certificate of Contribution.

This initial set deliberately exercises letters, tables, legal clauses, signatures, multi-page reports, commercial totals and certificate verification.

26 Template definition example

Illustrative service proposal definition:

type_code: DO-BIZ-PROPOSAL

family: BUSINESS

fields: client, contact, proposal_date, validity, project_title, executive_summary, scope, deliverables, assumptions, exclusions, timeline, fees, taxes, payment_terms, acceptance

workflow: author → reviewer → approver → issue

signature_roles: prepared_by, approved_by, client_acceptance

outputs: pdf, docx

confidentiality: internal/client

template_version: 1.0

27 Audit trail example

Example sequence for DO-BIZ-2026-0042:

09:14 — created by user A.

09:22 — client and scope updated.

09:40 — revision 1 generated.

10:05 — submitted for review.

10:21 — reviewer B requested pricing clarification.

10:37 — revision 2 created.

10:52 — reviewer B approved.

11:03 — approver C approved.

11:07 — final PDF generated; SHA-256 stored.

11:12 — authorized signer completed signature.

11:13 — document issued.

11:14 — authorized user downloaded artifact.

28 UX and interface architecture

Primary navigation should be operational: **Inbox · Documents · Templates · Clients · Projects · Certificates · Reports · Administration.**

The document workspace can use a split view: structured metadata/workflow on one side, document preview on the other.

Do not reproduce the public DailyOps website's decorative patterns. This is an operations application: state, responsibility, deadlines and evidence come first.

Every screen should make the next operational action obvious.

29 Governance

Each template has an owner responsible for content correctness and an authorized publisher responsible for releasing versions.

Document types define required fields, workflow, signature policy, retention, confidentiality and output rules.

Legal/formal template changes require elevated review and explicit versioning.

Numbering, certificate verification and audit configuration are protected administrative capabilities.

30 Launch checklist

Product: finalize document catalogue, workflow matrix, roles and permissions.

Design: finalize DailyOps document visual tokens, header/footer, typography, signature blocks and certificate system.

Engineering: initialize repository, schema, authentication, template engine, renderer, object storage and audit subsystem.

Content: build and validate the first 15–20 controlled templates against real DailyOps use cases.

Testing: test concurrent numbering, permissions, rejection, revision history, regenerated PDFs, signed artifact integrity and certificate revocation.

Operations: backup, retention, monitoring, recovery and administrator procedures.

Launch: issue new DailyOps documents through OpsDocs first; migrate historical documents progressively.

31 Definition of Done for v1

An authorized user can create a document from an approved type, complete structured fields, save and revise it, submit it through workflow, record review/approval/signature events, generate an immutable issued PDF with a unique reference, retrieve it later and inspect a complete audit timeline.

For certificates, v1 additionally generates a unique certificate ID and a verification endpoint that accurately reflects validity/revocation.

The system should make it difficult to accidentally produce an unofficial or structurally incorrect DailyOps document.

32 Recommended product position

Product: OpsDocs

Category: DailyOps Document Operations System

Core promise: Create. Control. Prove.

Internal definition: controlled system of record for creating, reviewing, approving, signing, issuing, verifying and archiving DailyOps documents.

OpsDocs should be a platform capability inside the DailyOps ecosystem and can later expose document services to other DailyOps systems.